13) Taban genişliği 8m , n=0.025, taban eğimi 0.0004 olan bir dikdörtgen kanaldan 2.5 derinliğindeki kesitten 20 m^3/s 'lik debi geçmektedir. 1 km ve 2 km ilerideki kesitte derinliğin ne olacağını bulunuz.

$$L = x_1 - x_2 = \frac{y_0}{y_1}\{(\eta_1 - \eta_2) - (1-\beta)[B(\eta_1) - B(\eta_2)]\}$$

$$\beta = \frac{C^2 J_0 B}{gP} \qquad \eta = \frac{y}{y_0} \qquad \eta_1 = \frac{y_1}{y_0} \qquad \eta_2 = \frac{y_2}{y_0}$$

ÇÖZÜM-13: $Q = \frac{1}{n} A R^{2/3} J_0^{1/2}$

$$20 = \frac{1}{0.025} \times 8y_0 \times \left(\frac{8y_0}{8+2y_0}\right)^{2/3} \times 0.0004^{1/2} \quad \Rightarrow \quad 3.125 = y_0 \times \left(\frac{8y_0}{8+2y_0}\right)^{2/3} \quad \Rightarrow \quad y_0 = 2.39\ \text{m}$$

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{20^2}{8^2 \times 9.81}} = 0.86\text{m} \qquad y_c < y_0 \quad \text{NR}$$

Üniform Akım için; $A_0 = 2.39 \times 8 = 19.12\ m^2$ $P_0 = 8+2\times2.39 = 12.78$ m $\Rightarrow R_0 = \frac{A_0}{P_0} = \frac{19.12}{12.78} = 1.496\,\text{m}$

$$C = \frac{R^{1/6}}{n} = \frac{1.496}{0.025} = 42.78\,\text{m}^{1/2}/\text{sn} \qquad \beta = \frac{C^2 J_0 B}{gP} = \frac{42.782 \times 0.0004 \times 8}{9.81 \times 12.78} = 0.0467 \qquad (B >> h) \quad \beta = \frac{J_0}{J_{kr}}$$

$$J_c = \frac{q^2 n^2}{R^{4/3} y_c^{\,2}} \quad \Rightarrow \quad R = \frac{0.86 \times 8}{8 + 2 \times 0.86} = 0.70\ \text{m}$$

$$J_c = \frac{2.5^2 \times 0.025^2}{0.70^{4/3} \times 0.86^2} = 0.0085 > J_0 \quad \text{Küçük eğimli} \qquad \beta = \frac{0.0004}{0.0085} = 0.0467$$

$$L = \frac{y_0}{J_0}\{(\eta_1 - \eta_2) - (1-\beta)[B(\eta_1) - B(\eta_2)]\}$$

$$\eta_1 = \frac{y_1}{y_0} = \frac{2.5}{2.39} = 1.046 \; f \Rightarrow \; B(\eta_1) = (1.046) = 0.963$$

$$\eta_2 = \frac{y_2}{y_0} = \frac{y_2}{2.39}$$

$$1000 = \frac{2.39}{0.0004}\left\{\left(1.046 - \frac{y_2}{2.39}\right) - (1-0.0467)\left[0.963 - B\left(\frac{y_2}{2.39}\right)\right]\right\}$$

$$-0.878 = -\frac{y_2}{2.39} - (0.9533)\left[0.963 - B\left(\frac{y_2}{2.39}\right)\right]$$

$$0.04 = -\frac{y_2}{2.39} + (0.9533)B\left(\frac{y_2}{2.39}\right) \;\Rightarrow\; 0.042 = -\frac{y_2}{2.28} + B\left(\frac{y_2}{2.39}\right) \;\Rightarrow\; y_2 = 2.449 \text{ m}$$

$$B\left(\frac{2.449}{2.39}\right) = B(1.025) = 1.1255$$

$$0.042 = -\frac{2.449}{2.28} + B(1.025)$$

$0.042 = -1.074 + 1.1255 \approx 0.05$

$y_2 = 2.449$ m L = 2 km ; $y_2 = 2.426$ m.

14) Bir nehirde q=1.8 m^2/s, C=45 $m^{1/2}/s$, $J_0=10^{-4}$'dür. Mansap x=0 kontrol kesitinde su seviyesi üniform akım derinliğinin 0.5 m üzerindedir. x≠0 farklı noktalar için seviyeleri hesaplayınız.

$$\Delta x = x_2 - x_1 = \frac{y_0}{J_0}\{(\eta_2\eta_1) - \gamma[\psi(\eta_2) - \psi(\eta_1)]\},\quad \gamma = 1 - \frac{C^2 J_0}{g},\quad \eta = \frac{y}{y_0},\quad J_{kr} = \frac{g}{C^2},\quad y_o = \sqrt[3]{\frac{q^2}{C^2 J_o}},$$

ψ(η):bresse fonk.

Bresse Fonk.

η	ψ(η)	H	ψ(η)	η	ψ(η)	η	ψ(η)
0.00	0.00	0.90	1.218	1.002	1.953	1.18	0.509
0.10	0.10	0.91	1.257	1.005	1.649	1.20	0.479
0.20	0.20	0.92	1.300	1.010	1.419	1.25	0.420
0.30	0.302	0.93	1.348	1.02	1.191	1.30	0.373
0.40	0.407	0.94	1.403	1.03	1.060	1.35	0.335
0.50	0.517	0.95	1.467	1.04	0.970	1.40	0.304
0.60	0.637	0.96	1.545	1.05	0.896	1.50	0.257
0.65	0.703	0.97	1.644	1.06	0.838	1.60	0.218
0.70	0.776	0.98	1.783	1.07	0.790	1.70	0.190
0.75	0.857	0.990	2.017	1.08	0.749	1.80	0.166
0.80	0.950	0.995	2.250	1.09	0.712	1.90	0.146
0.82	0.993	0.998	2.690	1.10	0.681	2.00	0.132
0.84	1.040	0.999	2.788	1.12	0.626	2.50	0.082
0.86	1.092	1.000	∞	1.14	0.580	3.00	0.055
0.88	1.151	1.001	2.184	1.16	0.541	more bigger	1/2 η^2

ÇÖZÜM-14: $y_0 = \sqrt[3]{\frac{q^2}{C^2 \times J_0}} = \sqrt[3]{\frac{1.8^2}{45^2 \times 10^{-4}}} = 2.52$ m

$B >> y$ $\quad y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{1.8^2}{9.81}} = 0.69$m $\quad y_0 > y_c$ NR.

$$J_c = \frac{g}{C^2} = \frac{9.81}{45^2} = 0.0048$$

$(B >> y)$ $J_0 << J_{kr}$ $y > y_o > y_{kr}$ M1

$$x_2 - x_1 = \frac{y_0}{J_0}\{(\eta_2 - \eta_1) - \gamma[\psi(\eta_1) - \psi(\eta_2)]\}$$

$$\gamma = 1 - \left(\frac{C^2 J_0}{g}\right) = 1 - (45^2 \times 10^{-4})/9.81 = 0.98 \cong 1.00$$

Sınır şartı x=0 $y_1 = y_0 + 0.5\ m = 2.52 + 0.5 = 3.02\ m$

$$\eta_1 = \frac{y_1}{y_0} = \frac{3.02}{2.52} = 1.20 \quad \eta_2 = 1.18 \quad y_2 = \eta_2\, y_0 = 1.18 \times 2.52 = 2.97\ m$$

i	η_i	ψ_i	$-x_i$(km)	y_i
0	1.20	0.479	0	3.02
1	1.18	0.509	1.26	2.97
2	1.16	0.541	2.57	2.92
3	1.14	0.580	4.06	2.87
4	1.12	0.626	5.72	2.82
5	1.10	0.681	7.61	2.77
6	1.08	0.749	9.83	2.72
7	1.06	0.838	12.57	2.67
8	1.04	0.970	16.40	2.62
9	1.02	1.191	42.13	2.57

Problem 8.14: In a river; $q=1.8\ m^2/sn$ $C=45\ m^{1/2}/sn$ $J_0=10^{-4}$. At the control section of river $x=0$ (downstream of the river), 0.5 m of water level is above the uniform water depth. Calculate the water levels for different sections ($x\neq0$).

$$\Delta x = x_2 - x_1 = \frac{y_0}{J_0}\{(\eta_2\eta_1) - \gamma[\psi(\eta_2) - \psi(\eta_1)]\}, \qquad \gamma = 1 - \frac{C^2 J_0}{g}, \qquad \eta = \frac{y}{y_0}, \qquad J_{kr} = \frac{g}{C^2},$$

$y_o = \sqrt[3]{\frac{q^2}{C^2 J_o}}$, $\psi(\eta)$: Bresse function

Bresse Function

η	ψ(η)	H	ψ(η)	η	ψ(η)	η	ψ(η)
0.00	0.00	0.90	1.218	1.002	1.953	1.18	0.509
0.10	0.10	0.91	1.257	1.005	1.649	1.20	0.479
0.20	0.20	0.92	1.300	1.010	1.419	1.25	0.420
0.30	0.302	0.93	1.348	1.02	1.191	1.30	0.373
0.40	0.407	0.94	1.403	1.03	1.060	1.35	0.335
0.50	0.517	0.95	1.467	1.04	0.970	1.40	0.304
0.60	0.637	0.96	1.545	1.05	0.896	1.50	0.257
0.65	0.703	0.97	1.644	1.06	0.838	1.60	0.218
0.70	0.776	0.98	1.783	1.07	0.790	1.70	0.190
0.75	0.857	0.990	2.017	1.08	0.749	1.80	0.166
0.80	0.950	0.995	2.250	1.09	0.712	1.90	0.146
0.82	0.993	0.998	2.690	1.10	0.681	2.00	0.132
0.84	1.040	0.999	2.788	1.12	0.626	2.50	0.082
0.86	1.092	1.000	∞	1.14	0.580	3.00	0.055
0.88	1.151	1.001	2.184	1.16	0.541	more bigger	$1/2\ \eta^2$

Solution :

$$y_0 = \sqrt[3]{\frac{q^2}{C^2 \times J_0}} = \sqrt[3]{\frac{1.8^2}{45^2 \times 10^{-4}}} = 2.52 \text{ m}$$

by assumption; $B >> y$

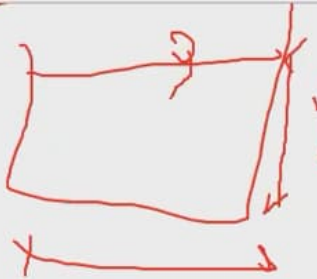
$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{1.8^2}{9.81}} = 0.69 \text{m}$$

Solution :

$$y_0 = \sqrt[3]{\frac{q^2}{C^2 \times J_0}} = \sqrt[3]{\frac{1.8^2}{45^2 \times 10^{-4}}} = 2.52\ \text{m}$$

by assumption; B >> y

$$y_c = \sqrt[3]{\frac{q^2}{g}} = \sqrt[3]{\frac{1.8^2}{9.81}} = 0.69\,\text{m}$$



$y_0 > y_c$ sub-critical flow

$$J_c = \frac{g}{C^2} = \frac{9.81}{45^2} = 0.0048$$

(B >> y)

$J_0 << J_{kr}$ $y > y_o > y_{kr}$

So water surface profile is M1 type

Using Bresse method;

$$x_2 - x_1 = \frac{y_0}{J_0}\{(\eta_2 - \eta_1) - \gamma[\psi(\eta_1) - \psi(\eta_2)]\}$$

$$\gamma = 1 - \left(\frac{C^2 J_0}{g}\right) = 1 - (45^2 \times 10^{-4})/9.81 = 0.98 \cong 1.00$$

at boundary condition x=0

$y_1 = y_0 + 0.5\ m = 2.52 + 0.5 = 3.02\ m$

$$\eta_1 = \frac{y_1}{y_0} = \frac{3.02}{2.52} = 1.20 \quad \eta_2 = 1.18$$

$y_2 = \eta_2\ y_0 = 1.18 \times 2.52 = 2.97\ m$

giving of η_i, ψ_i, x_i calculate y_i

Table p8.1a

i	η_i	ψ_i	$-x_i$(km)	y_i
0	1.20	0.479	0	3.02
1	1.18	0.509	1.26	2.97
2	1.16	0.541	2.57	2.92
3	1.14	0.580	4.06	2.87
4	1.12	0.626	5.72	2.82
5	1.10	0.681	7.61	2.77
6	1.08	0.749	9.83	2.72
7	1.06	0.838	12.57	2.67
8	1.04	0.970	16.40	2.62
9	1.02	1.191	42.13	2.57

Problem 8.15: A trapezoidal channel of base width 6.1 m and side slopes 1.5(horizontal)/1(vertical) is laid on a slope 0.001 and carries a discharge of 28.4 m^3/sn. The channel terminates in a free overflow and it is desired to explore the "drawdown" to the overflow and the extent to which velocities are thereby increased. For this it is required to compute and plot the flow profile upstream from the overflow over the region where the velocity is at least 10 percent greater than at uniform flow. The Manning n=0.025.

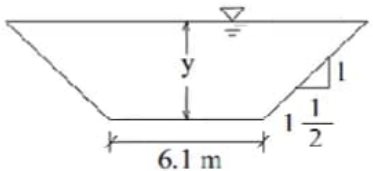


Figure 8.15

Solution: Problem will be solved to find the lengths from depths using direct step method.

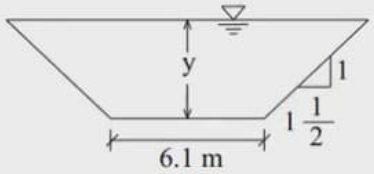


Figure p8.11a

$A = y(6.1+1.5\times y)$ $\quad P = 6.1+2y\sqrt{3.25} = 6.1+3.6y$

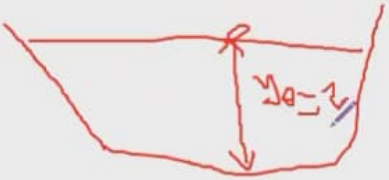
Uniform flow depth

$$Q = \frac{1}{n}A_0 R^{2/3} J_0^{1/2}$$

$$28.4 = \frac{1}{0.025}\times\left(\frac{y_0(6.1+1.5\times y_0)}{6.1+3.6y_0}\right)^{2/3}\times 0.001^{1/2}\times y_0(6.1+1.5\times y_0)$$

$$22.45 = \frac{(y_0(6.1+1.5\times y_0))^{5/3}}{(6.1+3.6y_0)^{2/3}}$$

by trial and error $y_0 = 2$ m



$A_0 = 2\times(6.1+1.5\times2) = 18.2\ m^2$

$V_0 = 1.56$ m/s

increasing the velocity by range of 10 % will add toV_0 , the velocity is V=1.716 m/sn

depth will be ten percent less than y_0.

y = 1.83 m (2/1.1=1.83)

Critical depth;

$$\frac{Q^2}{g} = \frac{A^3}{B} \rightarrow \frac{28.4^2}{9.81} = \frac{[y_c(6.1+1.5y_c)]^3}{[6.1+3y_c]}$$

Again by trial and error $y_c = 1.17$ m $\quad y < y_c$ Sub-critical flow

The range of depths to be covered therefore runs from $y_c = 1.17$ m at the overfall to about $y_0 = 1.83$ m ; the overflow itself is the control point from which the calculation must begin.

Table p8.2

1	2	3	4	5	6	7	8	9	10	11	12	13	14
y (m)	A (m^2)	P (m)	R (m)	C^2 (m/sn^2)	V (m/sn)	$V^2/2g$ (m)	E (m)	$J=(V^2/C^2R)$	$J_m=(V^2/C^2R)_m$	$(J_{0m}-J_m)=$ $J_m=(V^2/C^2R)_m$	ΔE (m)	Δx (m)	$X=\sum\Delta x$ (m)
1.17	9.24	10.33	0.89	1539	3.06	0.479	1.65	0.00683					0
									0.00572	-0.00472	0.026	-5.18	
1.31	10.58	10.82	0.98	1588	2.68	0.365	1.676	0.00460					5.18
									0.00412	-0.00312	0.035	-11.27	
1.4	11.50	11.16	1.03	1615	2.46	0.309	1.711	0.00364					16.45
									0.00327	-0.00227	0.046	-20.12	
1.49	12.44	11.46	1.09	1646	2.27	0.264	1.757	0.00290					36.57
									0.00262	-0.00162	0.053	-33.53	
1.58	13.42	11.79	1.14	1670	2.11	0.226	1.81	0.00234					70.10
									0.00213	-0.00113	0.060	-53.95	
1.67	14.44	12.13	1.19	1695	1.96	0.196	1.87	0.00191					124.05
									0.00177	-0.00077	0.057	-71.32	
1.76	15.28	12.41	1.23	1713	1.85	0.175	1.927	0.00163					195.37
									0.00151	-0.00051	0.058	-112.77	
1.85	16.16	12.68	1.27	1734	1.75	0.156	1.985	0.00139					308.14
									0.00134	-0.000340	0.025	67.05	
1.86	16.52	12.78	1.30	1743	1.71	0.148	2.01	0.00129					375.19

5. $C^2 = \frac{R^{1/3}}{n^2}$

6. $V = \frac{Q}{A}$

8. $E = y + \frac{V^2}{2g}$

13. $\Delta X = \frac{\Delta E}{\left(J_0 - \frac{V^2}{C^2 R}\right)_m} = \frac{12}{11}$

$\Delta E = E_{i+1} - E_i$

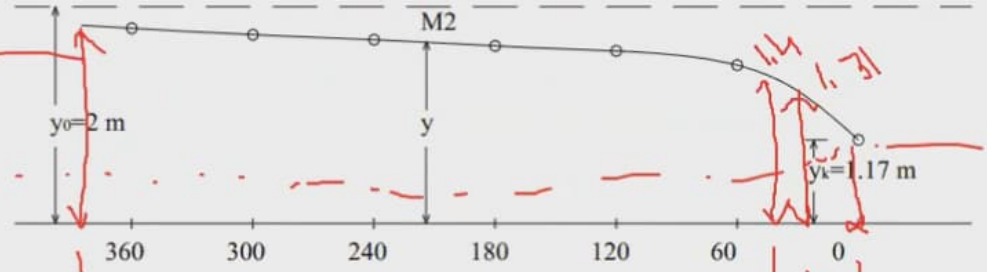


Figure p8.11b

$$x_{i+1} - x_i = \frac{E_{i+1} + E_i}{J_{om} - J_m}$$

MODEL TEORİSİ UYGULAMA SORU ÇÖZÜMLERİ

16)

a) Reynolds ve Froude Modellerini tanımlayınız.

b) Aynı bir modelde Reynolds ve Froude benzeşimlerinin aynı anda gerçekleşip gerçekleşmeyeceğini gerekçeli olarak açıklayınız.

ÇÖZÜM-16:

Reynolds modeli;

$$\mathrm{Re}_m = \mathrm{Re}_p \quad \Rightarrow \frac{\rho_m . V_m . L_m}{\mu_m .} = \frac{\rho_p . V_p . L_p}{\mu_p .} \quad \Rightarrow \frac{V_m}{V_p} = \frac{\mu_m}{\mu_p} . \frac{L_p}{L_m} . \frac{\rho_p}{\rho_m}$$

$$r_V = \frac{r_\mu}{r_L . r_\rho} \quad \Rightarrow r_\mu , r_\rho = 1 \quad \Rightarrow r_V = \frac{1}{r_L}$$

Froude Modeli;

$$Fr_m = Fr_p \quad \Rightarrow \frac{V_m^2}{g_m . L_m} = \frac{V_p^2}{g_p . L_p} \quad \Rightarrow \frac{V_m^2}{V_p^2} = \frac{L_m}{L_p} \frac{g_m}{g_p} \quad \Rightarrow r_V^2 = r_L$$

Aynı anda gerçekleşmesi için;

Aynı anda gerçekleşmesi için;

$$r_V = \frac{1}{r_L} = \sqrt{r_L}$$

sadece $r_L = 1$ iken, yani model ile prototipin aynı boyutlarda olmasıdır ki bu da istenileni gerçeklemez.

17) Laboratuarda kurulacak bir dalgakıran modeli dalga hareketine maruz bırakılarak, stabilitesi tahkik edilecektir. Model her biri 1kg ağırlığındaki beton bloklardan yapılmıştır ve dalga yüksekliği 0.3m geçmedikçe zarar görmemektedirler. Prototip de aynı tip beton bloklardan yapıldığına göre 6m yükseklikteki dalgaya mukavemet edebilmesi için bu beton blokların ağırlığı ne olmalıdır?

ÇÖZÜM-17:

G_m=1 kg $\quad$ G_p=?
H_m=0.3m $\quad$ H_p=6m

Prototipteki blokların G_p ağırlığı, dalganın hidrodinamik (F_p) etkisine karşı koyamadığında dalgakıran hasar görmeye başlar. Prototip ve Modeldeki olayların benzeşiminin gereği olarak;

$$\frac{G_m}{G_p}=\frac{F_m}{F_p} \qquad \frac{G_m}{G_p}=\frac{L_m^3}{L_p^3}\Rightarrow r_L^3$$

$$\frac{F_m}{F_p}=\frac{V_m^2 L_m^2}{V_p^2 L_p^2}\Rightarrow r_v^2 r_L^2$$

$$r_L^3=r_v^2 r_L^2 \quad \Rightarrow r_L=r_V^2$$

Froude Modeli;

$$Fr_m=Fr_p \qquad \Rightarrow \frac{V_m^2}{g_m H_m}=\frac{V_p^2}{g_p H_p} \qquad \Rightarrow \frac{V_m^2}{V_p^2}=\frac{H_m}{H_p}\frac{g_m}{g_p} \qquad \Rightarrow r_V^2=r_H$$

$$\boxed{r_L = r_V^2 = r_H}$$

$$r_H = \frac{0.3}{6} = \frac{1}{20} = r_L$$

$$\frac{G_m}{G_p} = r_L^3 = \left(\frac{1}{20}\right)^3 = \frac{1}{8000}$$

G_m=1kg $\Rightarrow$ G_p=8000kg=8t

18) Bir gelgit hareketini incelemek üzere planda 1/500 ve kesitte 1/50 ölçekli model tesis çizilmiştir.

a) Prototipte periyodu 12 saat 15 dakika olan bir gelgit hareketinin modeldeki periyodu ne olmalıdır?

b) Planda 1/50000 ve kesitte 1/500 ölçekli model için periyot ne olur?

ÇÖZÜM-18:

Yatay $\Rightarrow r_L = 1/500$

Düşey $\Rightarrow r_H = 1/50$ $\Rightarrow$ distorsiyonlu model.

Ağırlık kuvvetleri hakim olduğu için Froude Benzeşimi sağlanmalıdır.

$$\text{a) } r_{Fr} = 1 = \frac{Fr_m}{Fr_p} \Rightarrow \frac{V_m^2}{g.H_m} = \frac{V_p^2}{g.H_p} \Rightarrow \frac{V_m^2}{V_p^2} = \frac{H_m}{H_p} \Rightarrow r_V^2 = r_H = \frac{1}{50} \Rightarrow r_V = \sqrt{\frac{1}{50}}$$

$$r_T = \frac{T_m}{T_p} = \frac{L_m / V_m}{L_p / V_p} = \frac{L_m}{L_p}.\frac{V_p}{V_m} \Rightarrow \frac{r_L}{r_V}$$

$$r_T = \frac{r_L}{r_V} = \frac{1/500}{\sqrt{1/50}} = \frac{T_m}{T_p}.$$

$$T_m = T_p \cdot \frac{1/500}{\sqrt{1/50}} = 735 x \frac{1/500}{\sqrt{1/50}} = 10.4\, dk$$

b) $T_m = 735. \dfrac{1/50000}{\sqrt{1/500}} = 0.33dk$